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**B.TECH
(SEMVII) THEORY EXAMINATION 2022-23
PLANT DESIGN AND ECONOMICS**

Time: 3 Hours

Total Marks: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.
2. Any special paper specific instruction.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- (a) Define and explain cost index. Also, state its usefulness.
- (b) List out names of different types of process flow diagrams.
- (c) Define: (i) fixed capital (ii) working capital.
- (d) List out name of any three methods for profitability analysis.
- (e) Enlist any six factors for selection of location of a chemical process unit.
- (f) What is the role of chemical engineer in a chemical process industry?
- (g) The cost of a drum dryer is Rs. 10 lakhs. What will be the cost of a drum dryer with double the surface area in lakhs of rupees?
- (h) With suitable example write in brief about salvage value.
- (i) Define Order of Magnitude of Cost Estimation.
- (j) Give the significance of Break Even Point.

SECTION B

2. Attempt any three of the following: 10 x 3 = 30

- (a) Explain selection of material handling equipments in chemical plant.
- (b) Draw a generalized schematic flow diagram of design and development stages of an engineering project.
- (c) Enlist various factors to be considered for comparison of different processes. Discuss any two of them in detail.
- (d) Explain 'sixth-tenth factor rule'.
A heat exchanger of area 10 m² costed Rs 50,000 in 2009. What is the estimated cost of a 15 m² heat exchanger in 2014? Assume that the cost index in 2009 was 270 and in 2014 it is 320. Equipment cost-vs-capacity exponent is 0.6.
- (e) Explain about various utilities in chemical process industries.

SECTION C

3. Attempt any one part of the following: 10 x 1 = 10

- (a) State the preliminary specifications for equipments. Prepare the specification sheet for a heat exchanger.
- (b) A process plant has a life of 7 years and its salvage value is 30 %. For what minimum fixed – percentage factor will the depreciation amount for the second year, calculated by declining balance method be equal to that calculated by the straight line depreciation method?

4. Attempt any *one* part of the following: 10 x 1 = 10
- (a) What is payout period? Explain how you will compute payout period in years, with and without interest charge.
 - (b) The original value of a piece of equipment is \$22,000, completely installed and ready for use. Its salvage value is estimated to be \$2000 at the end of a service life estimated to be 10 years. Determine the asset (or book) value of the equipment at the end of 5 years using straight line and textbook declining balance method.
5. Attempt any *one* part of the following: 10 x 1 = 10
- (a) State and explain various factors for techno-economic feasibility survey of a Chemical process plant.
 - (b) Define pilot plant. What is the importance of laboratory development of pilot plant?
6. Attempt any *one* part of the following: 10 x 1 = 10
- (a) Explain fault tree analysis technique for hazard assessment.
 - (b) A proposed manufacturing plant requires an initial fixed capital investment of \$ 900,000 and \$ 100,000 of working capital. It is estimated that the annual income will be \$ 800,000 and the annual expenses including depreciation will be \$ 520,000 before income taxes. A minimum annual return of 15% before income tax is required before the investment would be worthwhile. Income taxes amount to 34 percent of all pre-tax profits. Determine the annual percent return on total investment before and after income taxes.
7. Attempt any *one* part of the following: 10 x 1 = 10
- (a) With example explain practical factors in alternative investment and replacement studies.
 - (b) A reactor needs to be lined with a corrosion resistant lining. One type of lining cost Rs.5 lakhs and is expected to last for 2 years. Another type of lining lasts for 3 years. If both choices have to be equally economical, with the effective interest rate being 18%, compounded annually, determine the price for the second type of lining ?